WEEK-6

Joint Continous Random Variable

↳ involves 2 variables (X, Y) defined on the same sample space.

- Key properties; → Joint density of X and Y
  - $f_{XY}(x,y) \geq 0$
  - Total probability = 1:

$$\iint_{\mathbb{R}^2} f_{XY}(x,y)\, dx\, dy = 1$$

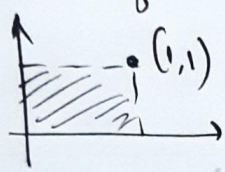
- Interpretation:
- Probability = area under surface

Ex: let X and Y have joint density:

$$f_{XY}(x,y) = \begin{cases} 1 & 0 < x < 1,\ 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

Sol.

$$\int_0^1 \int_0^1 1 \cdot dx\, dy = \int_0^1 [x]_0^1\, dy \Rightarrow [y]_0^1 \Rightarrow 1$$



Uniform Density

- Fix some region D in $\mathbb{R}^2$ with total area $|D|$.

We say that $(X,Y) \sim \text{Uniform}(D)$ if they have the joint density

$$f_{XY}(x,y) = \begin{cases} \frac{1}{|D|} & (x,y) \in D, \\ 0 & \text{otherwise} \end{cases}$$

$$f(x,y) = \frac{1}{|D|} \rightarrow \text{total area of region}$$

- For any sub-region A:

$$P(A) = \frac{\text{Area of } A}{\text{Area of } D}$$

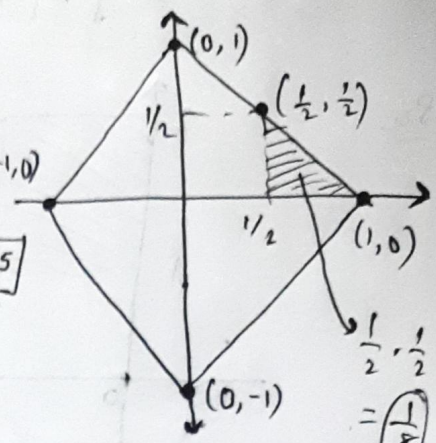
Q. ① Geometric Probability $\boxed{|x| + |y| \le 1}$

Find $P\left(\frac{1}{2} < x < 1,\ y > 0\right)$.

Sol.

$$\frac{\text{Req. Area}}{\text{Total Area}} = \frac{1/8}{2/1} = \frac{1}{8}\cdot\frac{1}{2}$$

$$\Rightarrow \frac{1}{16} = \boxed{0.625}$$



$\frac{1}{2}\cdot\frac{1}{2}\cdot\frac{1}{2} = \frac{1}{8}$

$|x| + |y| = 1$

$x + y = 1 \Rightarrow \frac{1}{2} + y = 1 \Rightarrow y = 1 - \frac{1}{2} = \frac{1}{2}$

Q ② Integral Method

let X and Y be continuous random variables with joint density

$$f_{XY}(x,y) = \begin{cases} 6x^2y & \text{for } 0<x<1,\ 0<y<1 \\ 0 & \text{otherwise} \end{cases}$$

① Find $P\left(0 < X < \frac{1}{2},\ 0 < Y < \frac{1}{2}\right)$.

Sol.

$$P = \frac{\text{Req. Area}}{\text{Total Area}} = \frac{\int_0^{1/2}\int_0^{1/2} 6x^2y\,.\,dx\,dy}{\int_0^1\int_0^1 6x^2y\,.\,dx\,dy} \Rightarrow \frac{1}{32} = 0.031$$

Marginal Density

Suppose (X,Y) have joint density $f_{XY}(x,y)$. Then,

- X has the marginal density $f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x,y)\,dy$
- Y has the marginal density $f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x,y)\,dx$

Q. Consider the joint density $f_{XY}(x,y) = \begin{cases} x+y, & 0<x,y<1 \\ 0, & \text{otherwise} \end{cases}$

Find the marginals.

Sol.

$$f_X(x) = \int_{y=0}^{1}(x+y)\,dy = x\int_0^1 (1)\,dy + \int_0^1 y\,.\,dy = x\,[y]_0^1 + \left[\frac{y^2}{2}\right]_0^1 = x + \frac{1}{2}$$

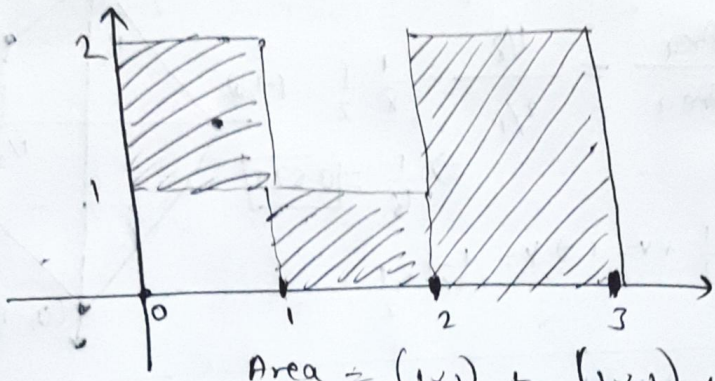
$$f_Y(y) = \int_0^1 (x+y)\,dx = \int_0^1 x\,.\,dx + y\int_0^1 1\,.\,dx = \left[\frac{x^2}{2}\right]_0^1 + y\,[x]_0^1$$

$$\Rightarrow \frac{1}{2} + y$$

Q. Let $(X,Y) \sim \text{Uniform}(D)$, where

$$D := \underbrace{[0,1]\times[1,2]} \cup \underbrace{[1,2]\times[0,1]} \cup \underbrace{[2,3]\times[0,2]}.$$

Sol.



$$\text{Area} = (1\times1) + (1\times1) + (1\times2) = 4$$

- $f_{XY}(x,y) = \frac{1}{|D|} = \frac{1}{4}$

- for $0<x<1$: x in Interval $[0,1]\times[1,2]$

  $1<y<2$: $f_X(x) = \int_1^2 \frac{1}{4}\,dy = \frac{1}{4}[y]_1^2 = \frac{1}{4}$

- for $1<x<2$:

  is covered in both $[0,1]\times[1,2]$ and $[1,2]\times[0,1]$

  $$f_X(x) = \int_0^1 \frac{1}{4}\,dy + \int_1^2 \frac{1}{4}\,dy \Rightarrow \frac{1}{2}$$

- for $2<x<3$: is the in interval $[2,3]\times[0,2]$

  $$f_X(x) = \int_0^2 \frac{1}{4}\,dy = \frac{1}{4}[2-0] = \frac{1}{2}$$

Ans. $f_X(x) = \begin{cases} \frac{1}{4} & \text{for } 0<x<2 \\ \frac{1}{2} & \text{for } 2<x<3 \end{cases}$

Q. Let X and Y be continuous random variables with joint density function

$$f_{XY}(x,y) = \begin{cases} \frac{1}{4} & \text{for } 0 < x < \frac{y}{2},\ 0 < y < 4 \\ 0 & \text{otherwise} \end{cases}$$

⇒ find the marginal density of X.

$$f_X(x) = \int_{2x}^{4} \frac{1}{4} \cdot dy \Rightarrow \frac{1}{4}[y]_{2x}^{4} \Rightarrow \frac{1}{4}[4-2x]$$

$$\Rightarrow 1 - \frac{x}{2}$$

for $0 < x < 2$

⇒ find the marginal density of Y.

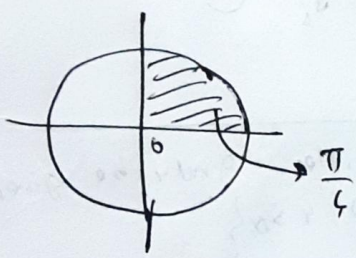
$$f_Y(y) = \int_0^{y/2} \frac{1}{4} \cdot dx = \frac{1}{4}[x]_0^{y/2} \Rightarrow \frac{1}{4}\left[\frac{y}{2} - 0\right] = \frac{y}{8}$$

for $0 < y < 4$

Q. ~~Let~~ Let $(X,Y) \sim \text{Uniform}(D)$, where $D := \{x^2 + y^2 \le 1,\ x > 0,\ y > 0\}$.

Sol. $x^2 + y^2 \le 1$

$(x-0)^2 + (y-0)^2 \le (1)^2$



$$f_{XY}(x,y) = \frac{1}{|D|} = \frac{1}{\frac{\pi}{4}} = \frac{4}{\pi}$$

$$f_X(x) = \int_0^{\sqrt{1-x^2}} \frac{4}{\pi}\, dy = \frac{4}{\pi}\int_0^{\sqrt{1-x^2}} dy = \frac{4}{\pi}[y]_0^{\sqrt{1-x^2}}$$

$$\Rightarrow \frac{4}{\pi}\left[\sqrt{1-x^2} - 0\right]$$

$$\Rightarrow \frac{4}{\pi}\left[\sqrt{1-x^2}\right]$$

Independence of multiple Random variable

$$\underbrace{f_{XY}(x,y)}_{\text{joint density of } (X,Y)} = \underbrace{f_X(x)}_{} \cdot \underbrace{f_Y(y)}_{}$$

marginal densities of X and Y respectively

Q.1. Let $X \sim Exp(2)$ and $Y = Exp(4)$ be independent.

Sol.

$$f_X = \lambda e^{-\lambda x}$$

$$f_Y = \lambda e^{-\lambda y}$$

$$f_{XY}(x,y) = f_X(x) \cdot f_Y(y) = 2e^{-2x} \cdot 4e^{-4y}$$

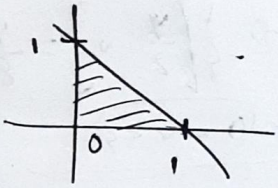
$$\Rightarrow 8e^{-2(x+2y)}$$

for $x \geq 0, y \geq 0$

Q.2 Let $(X,Y) \sim$ Uniform (D), where $D := \{(x,y): x+y<1, x>0, y>0$

Are X and Y independent?

Sol.



$$f_{XY}(x,y) = f_X(x) \cdot f_Y(y)$$

$$\frac{1}{|D|} = \frac{1}{(1/2)} \Rightarrow 2$$

$$f_X(x) = \int_0^1 2\, dy = 2[y]_0^1 = 2$$

$$f_Y(y) = \int_0^1 2 \cdot dx = 2[x]_0^1 = 2$$

$2 \neq 2 \cdot 2$ **No**

Q. Let the joint PDF of 2 random variables X and Y be given by

$$f_{XY}(x,y) = \begin{cases} e^{-(x+y)} & \text{for } x>0, y>0 \\ 0 & \text{otherwise} \end{cases}$$

① Check if $f_{XY}(x,y) = f_X(x) \cdot f_Y(y)$

Sol.

$$f_X(x) = \int_0^\infty e^{-(x+y)} \cdot dy = e^{-x}$$

$$f_Y(y) = \int_0^\infty e^{-(x+y)} dy = e^{-y}$$

$$f_{XY}(x,y) = e^{-(x+y)} \quad \checkmark \quad e^{-x} \cdot e^{-y} = e^{-(x+y)} \checkmark \text{ true}$$

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② check if $P(X<Y) = \frac{1}{2}$

$$\int_0^\infty \int_0^y f_{XY}(x,y)\, dx \cdot dy$$

$$\Rightarrow \int_0^\infty \int_0^y e^{-(x+y)} \cdot dx \cdot dy = \frac{1}{2}$$

check if $P(X<2) = 1-e^{-2}$

$$\Rightarrow \int_0^2 f_X(x)\, dx$$

$$\Rightarrow \int_0^2 e^{-x} dx$$

$$\Rightarrow \boxed{1 - e^{-2}}$$

Conditional Density of Multiple Continuous Random Variable

$$f_{Y|X=a}(y) = \frac{f_{XY}(a,y)}{f_X(a)}$$

$f_{XY}(a,y) \rightarrow$ Joint Density

$f_X(a) \rightarrow$ Marginal Density

Q. Let the Joint PDF of 2 random variables x and y be given by

$$f_{XY}(x,y) = \begin{cases} \frac{x(1+2y)}{4} & \text{for } 0<x<2,\ 0<y<1 \\ 0 & \text{otherwise} \end{cases}$$

Calculate $P\left(\frac{1}{8} < x < \frac{1}{4} \,\middle|\, y = \frac{1}{4}\right)$.

Sol. ~~$f_{XY}(x,y) = x$~~

(1) Use Conditional Prob. formula:

$$P(A|Y=y) = \frac{\int_A f_{XY}(x,y)\,dx}{f_Y(y)}$$

So here:

$$P\left(\frac{1}{8} < x < \frac{1}{4} \,\middle|\, y = \frac{1}{4}\right) = \frac{\int_{1/8}^{1/4} f(x, 1/4)\,dx}{f_Y(1/4)}$$

(2) $f\left(x, \frac{1}{4}\right) = \frac{x(1+2(1/4))}{4} = \frac{3x}{8}$

(3) $\int_{1/8}^{1/4} \frac{3x}{8}\,dx \Rightarrow \frac{3}{8}\int_{1/8}^{1/4} x\,dx = \frac{3}{8}\left[\frac{x^2}{2}\right]_{1/8}^{1/4}$

$\Rightarrow \frac{9}{1024}$

(4) $f_Y(y) = \int_0^2 \frac{x(1+2y)}{4}\,dx = \frac{1+2y}{4}\int_0^2 x\,dx$

$\Rightarrow \frac{1+2y}{4}\cdot\left[\frac{x^2}{2}\right]_0^2 = \frac{1+2y}{2}$

$y = \frac{1}{4} \Rightarrow f_Y(1/4) = \frac{3}{4}$

(5) $\text{ans} = \frac{9/1024}{3/4} = \frac{3}{256} \approx 0.01$